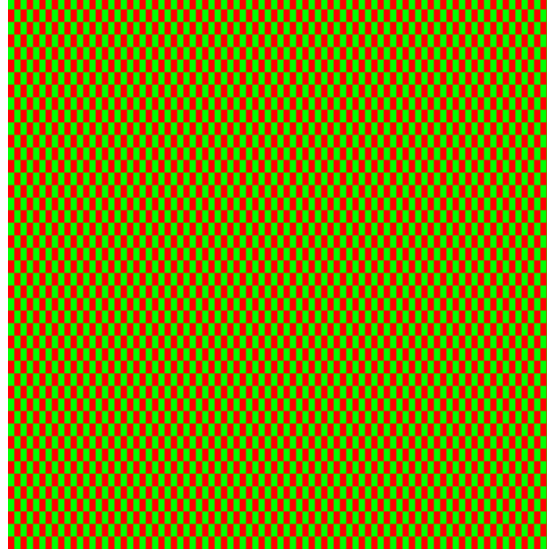


CS 6959 — Project 3

Ray Tracing Kernels

A. Pixel-tile mapping



Page 5 kernel, unmodified mapping; 350×350 framebuffer.

B. Primary ray visibility



(a) `--scene-name=triangle` · 350×350



(b) `--scene-name=teapot` · 350×350

Each 32-thread tile processes a 4×8 pixel region. Rounded-up tile counts and bounds checks handle partial edge tiles without mixing distant pixels. One camera ray is traced per valid pixel using `_traceray<0x0U>(0, ray, hit)`; hits are red and misses dark gray.

Simulator termination fixes prevent repeated halt counting and drain pending framebuffer stores. Both visibility images match a CPU brute-force triangle-intersection reference at every pixel. Submitted separately: `main.cpp` and one `trax_log.txt` containing both Part B runs.